

Current and Future Technologies

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OUTLINE



- Executive Summary
- Introduction
- Methodology
- Results
 - Visualization – Charts
 - Dashboard
- Discussion
 - Findings & Implications
- Conclusion
- Appendix



EXECUTIVE SUMMARY



- Current Technology Usage
- Future Technology Trends
 - JavaScript & SQL are highly desired for future use.
 - AWS is the most desired platform for future use.
 - React & Node.js are the most popular web frameworks for the future.
- Demographics Information
- Respondent Distribution by Age
- Respondent Count by Education Level

INTRODUCTION



About:

- Analyzing Current & Future Technology Trends

Purpose of Report:

- Identify the most widely used programming languages, databases, platforms, and web frameworks
- Identify technologies developers are most interested in using in the future

Target Audience:

- Developers and Technology Professionals

Value of Analysis:

- Helps developers understand which technologies are currently widely used
- Helps identify skills and technologies that may be valuable to learn for the future



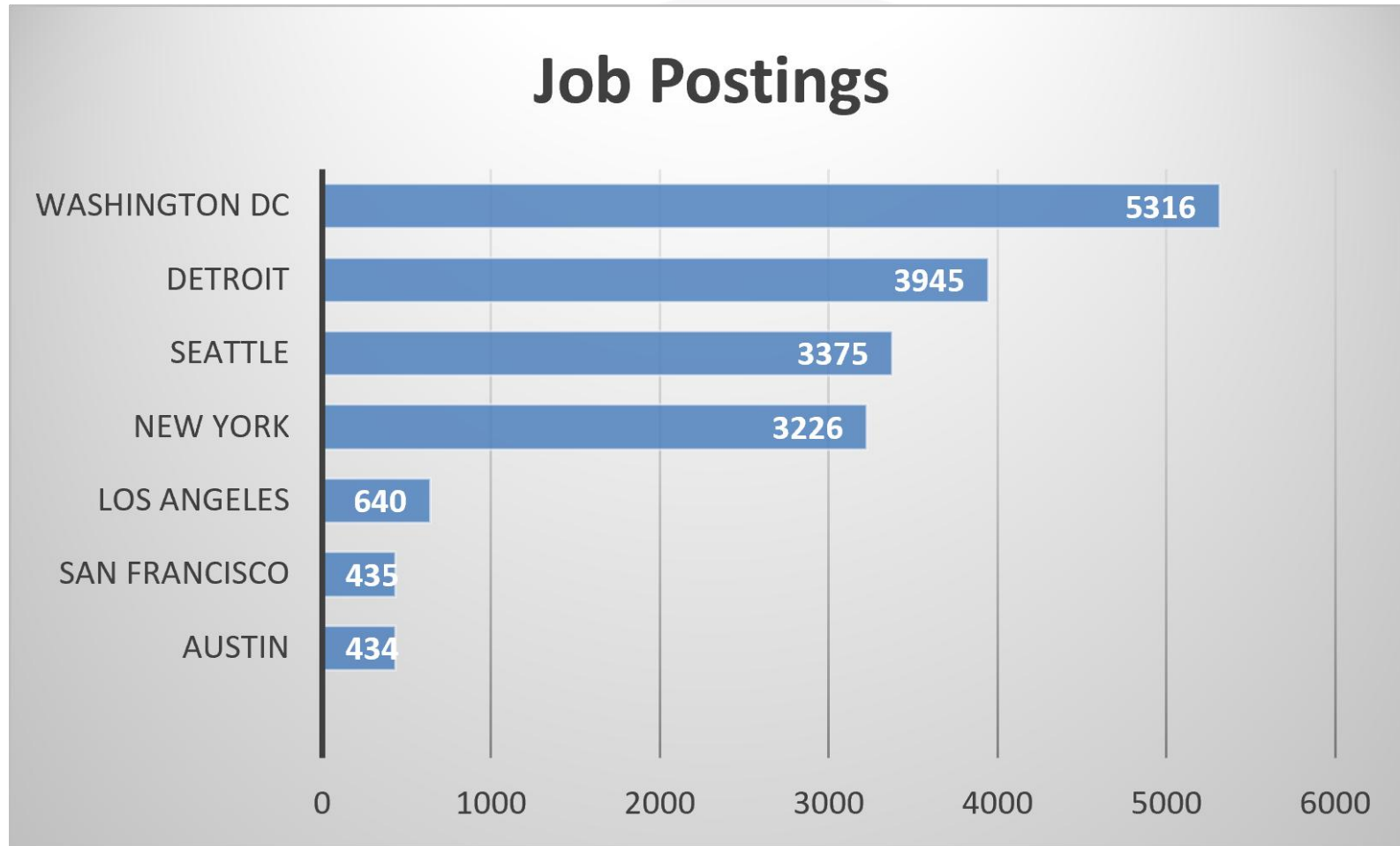
METHODOLOGY



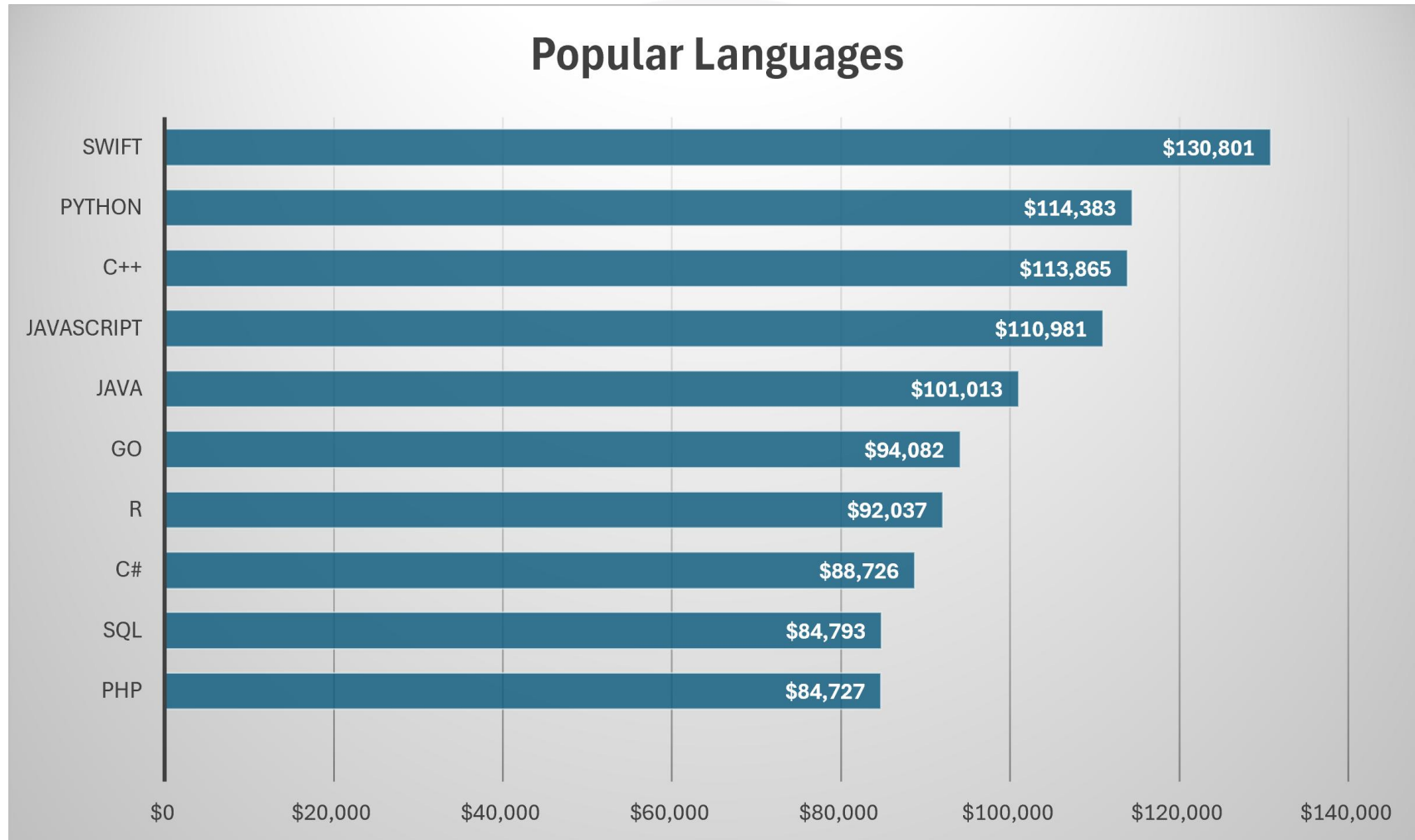
- Data Source: Data was provided in CSV format
- Data Collection: Data was downloaded and used for analysis
- Data Wrangling:
 - Used Python (Pandas) to select the relevant columns
 - Split values separated by ; into individual values
 - Used `.explode()` to place each language on a new row
 - Used `.value_counts()` to count each language and sort from highest to lowest
 - Used `.head(10)` to select the top 10 languages
- Data Visualization:
 - Used IBM Cognos Analytics to create visualizations and dashboards.
 - Used charts such as bar charts, pie charts, treemaps, bubble charts to present the data.



JOB POSTINGS

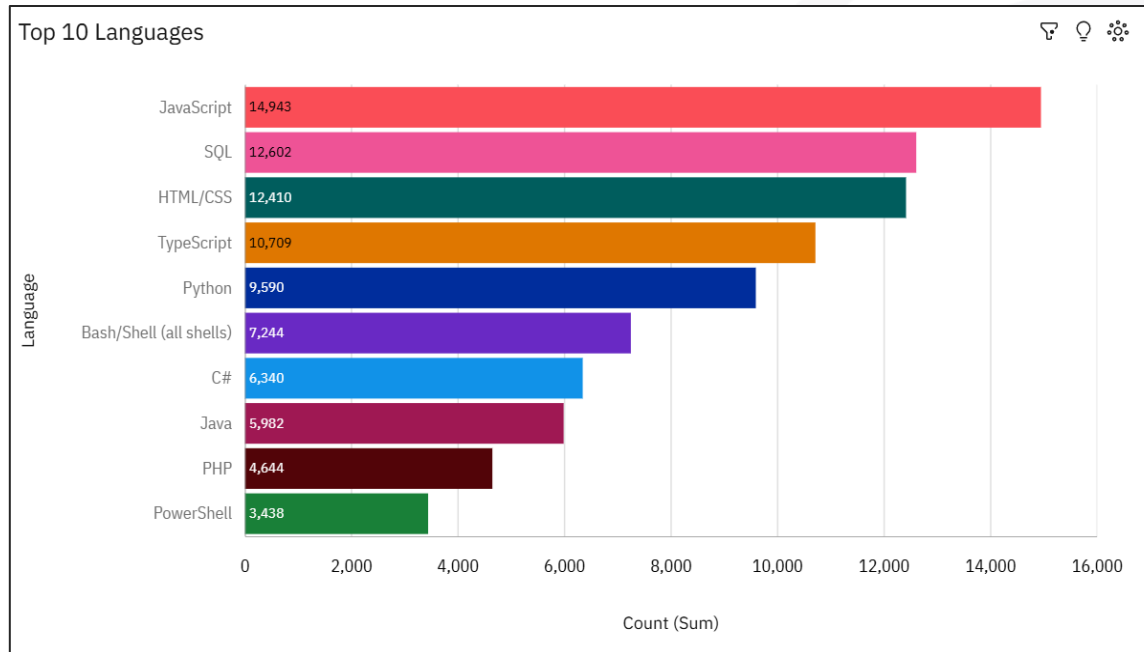


POPULAR LANGUAGES

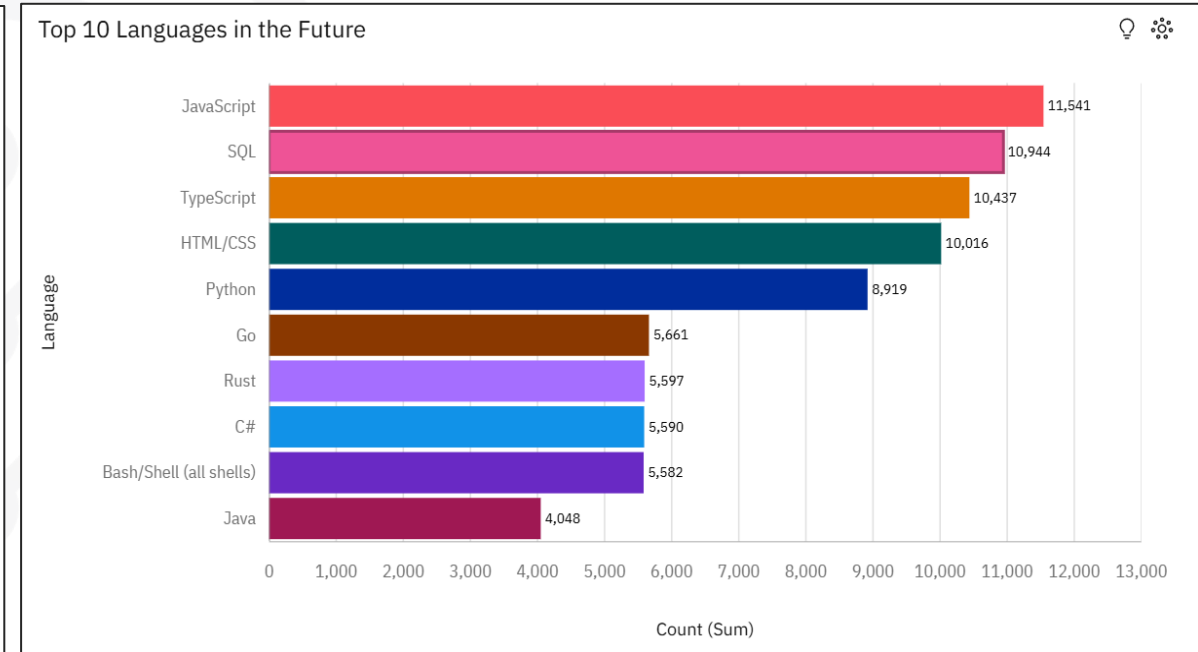


PROGRAMMING LANGUAGE TRENDS

Current Year:



Next Year:



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings:

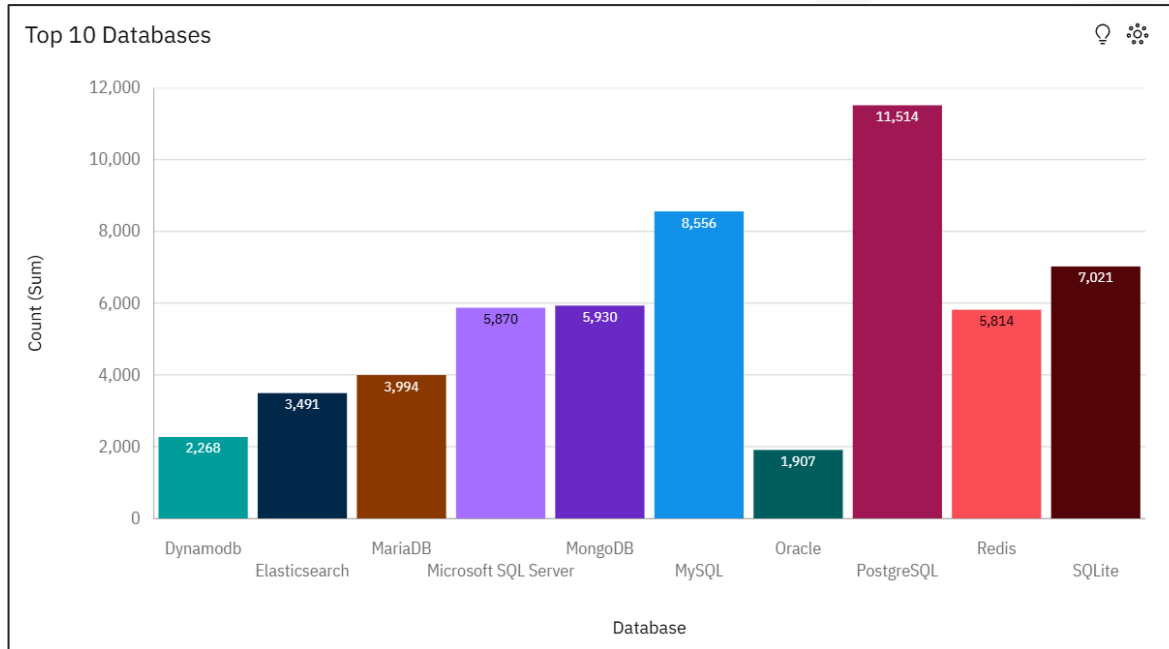
- JavaScript, SQL, and HTML/CSS are the top 3 currently used languages.
- Future interest is lower for Java, Python, C#, and Bash/Shell compared with current usage.
- Go and Rust enter the future top 10, while PHP and PowerShell drop out.

Implications

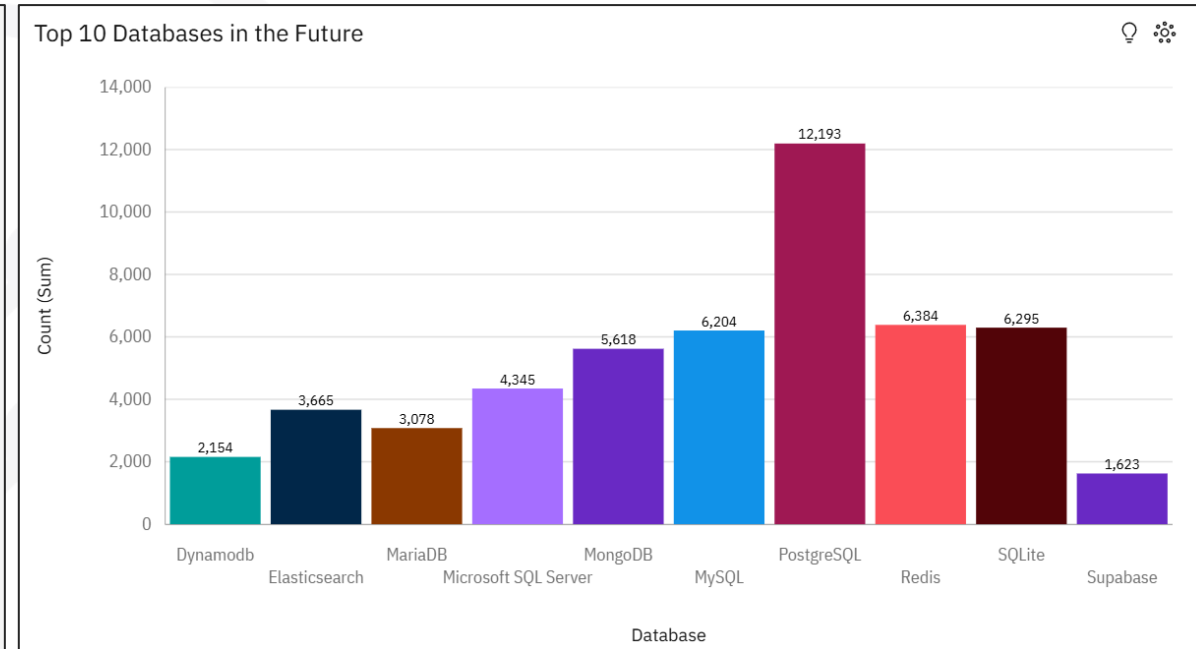
- Web development technologies continue to be widely used.
- Growing interest in Go and Rust suggests they may become important for developers in the future.
- Developers may need to keep their programming skills up to date as technology preferences change.

DATABASE TRENDS

Current Year:



Next Year:



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- PostgreSQL is the most widely used database currently and also has the highest future interest.
- MySQL is currently the second most widely used database, but future interest is lower.
- Oracle appears in the current top 10 but drops out of the future top 10.

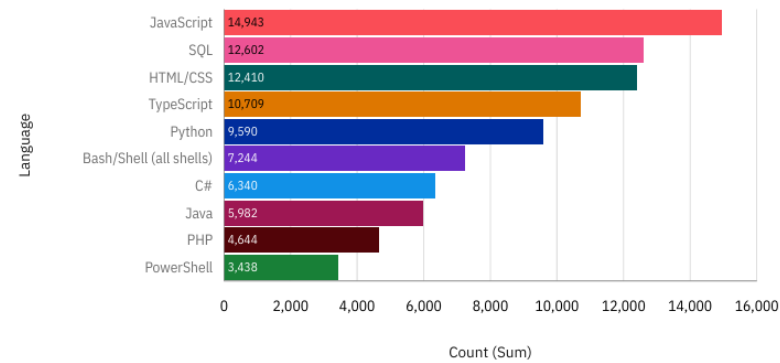
Implications

- PostgreSQL is likely to remain an important database for developers.
- Future interest in MySQL may be lower compared with its current usage.
- The appearance of Supabase in the future top 10 suggests growing interest in newer database technologies.

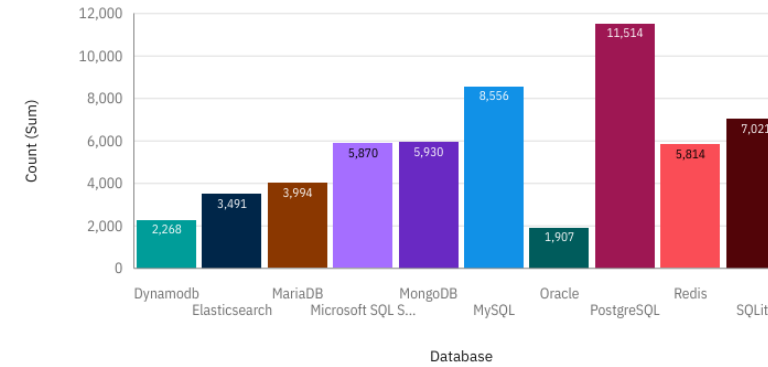
DASHBOARD TAB 1: Current Technology Usage

Current Technology Usage

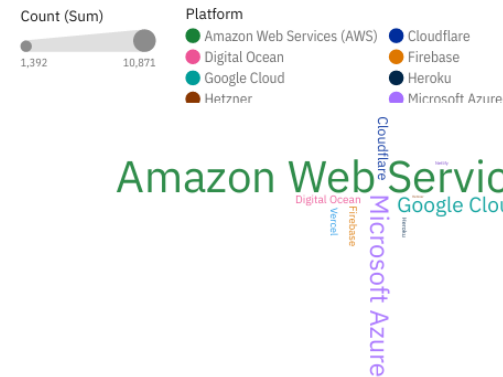
Top 10 Languages



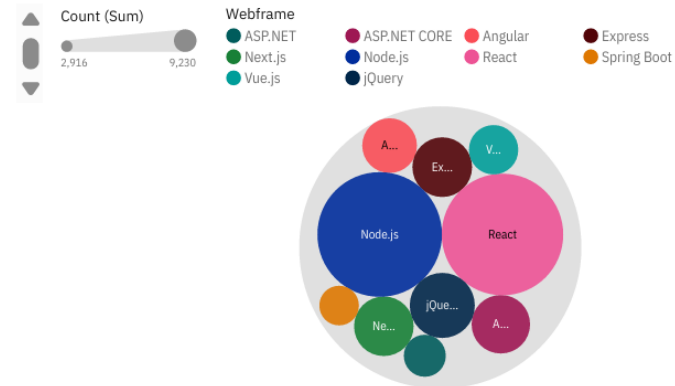
Top 10 Databases



Top 10 Platforms



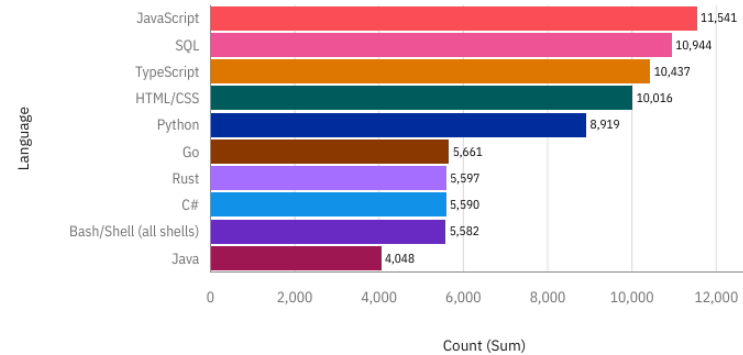
Top 10 Webframes



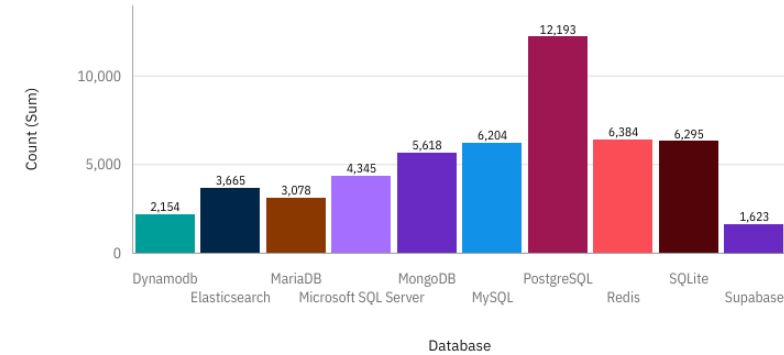
DASHBOARD TAB 2: Future Technology Trends

Future Technology Trend

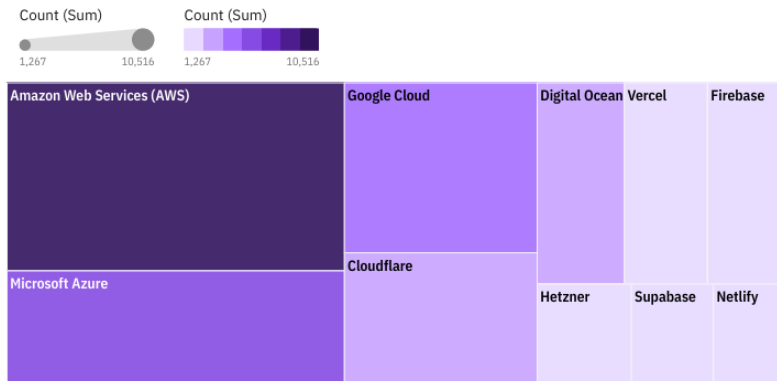
Top 10 Languages in the Future



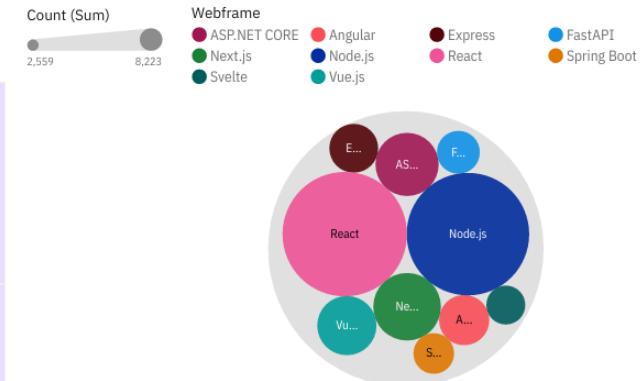
Top 10 Databases in the Future



Top 10 Platforms in the Future



Top 10 Webframes in the Future



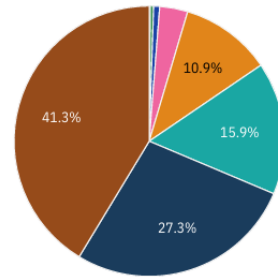
DASHBOARD TAB 3: Demographics

Demographics

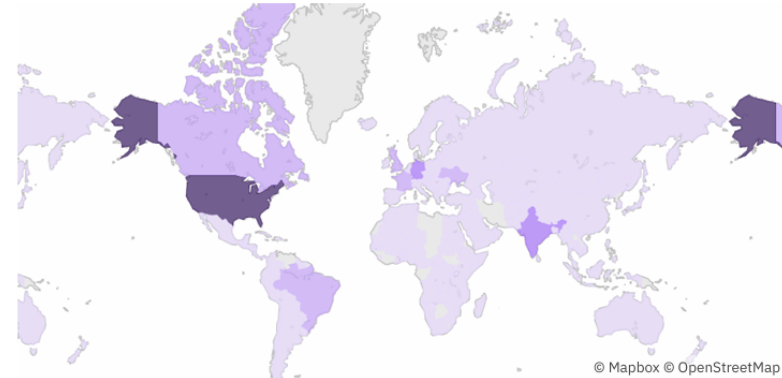
Respondent Distribution by Age

Age

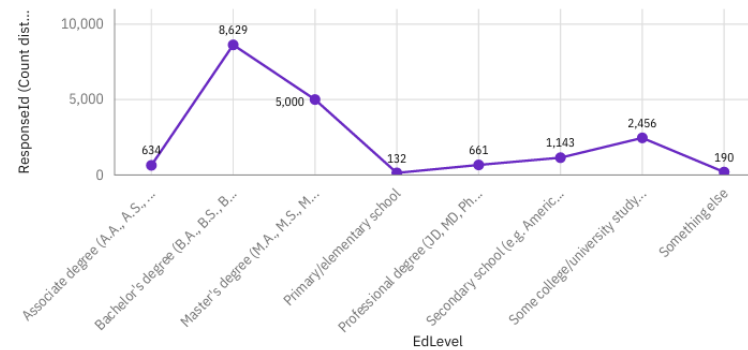
- Prefer not to say
- 18-24 years old
- 25-34 years old
- 35-44 years old
- 45-54 years old
- 55-64 years old
- 65 years or older
- Under 18 years old



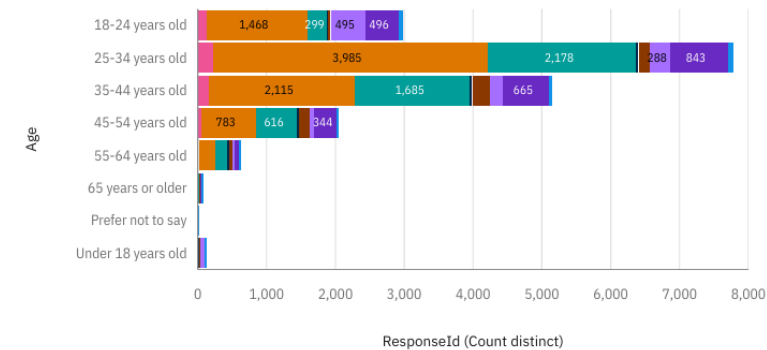
Respondent Count by Country



Respondent Distribution by Formal Education Level



Respondent Count by Age, classified by Education Level



OVERALL FINDINGS & IMPLICATIONS

Findings

- Current Technology - JavaScript, PostgreSQL, AWS, and React/Node.js are among the leading technologies currently used by respondents.
- Future Technology - JavaScript, PostgreSQL, AWS, React, and Node.js continue to show strong interest for future use.
- Demographics - Most respondents are young adults, with the 25–34 age group representing the largest share (41.3%), and bachelor's degrees being the most common education level.

Implications

- Leading technologies remain as important skills for developers.
- Developers may benefit from focusing on technologies with strong current and future interest.
- Most of the data comes from 25–34-year-olds, and a Bachelor's degree is the most common education level.

CONCLUSION



- Current and Future Technology Trends
- Emerging Technologies
- Future Skills
- Demographics Overview

